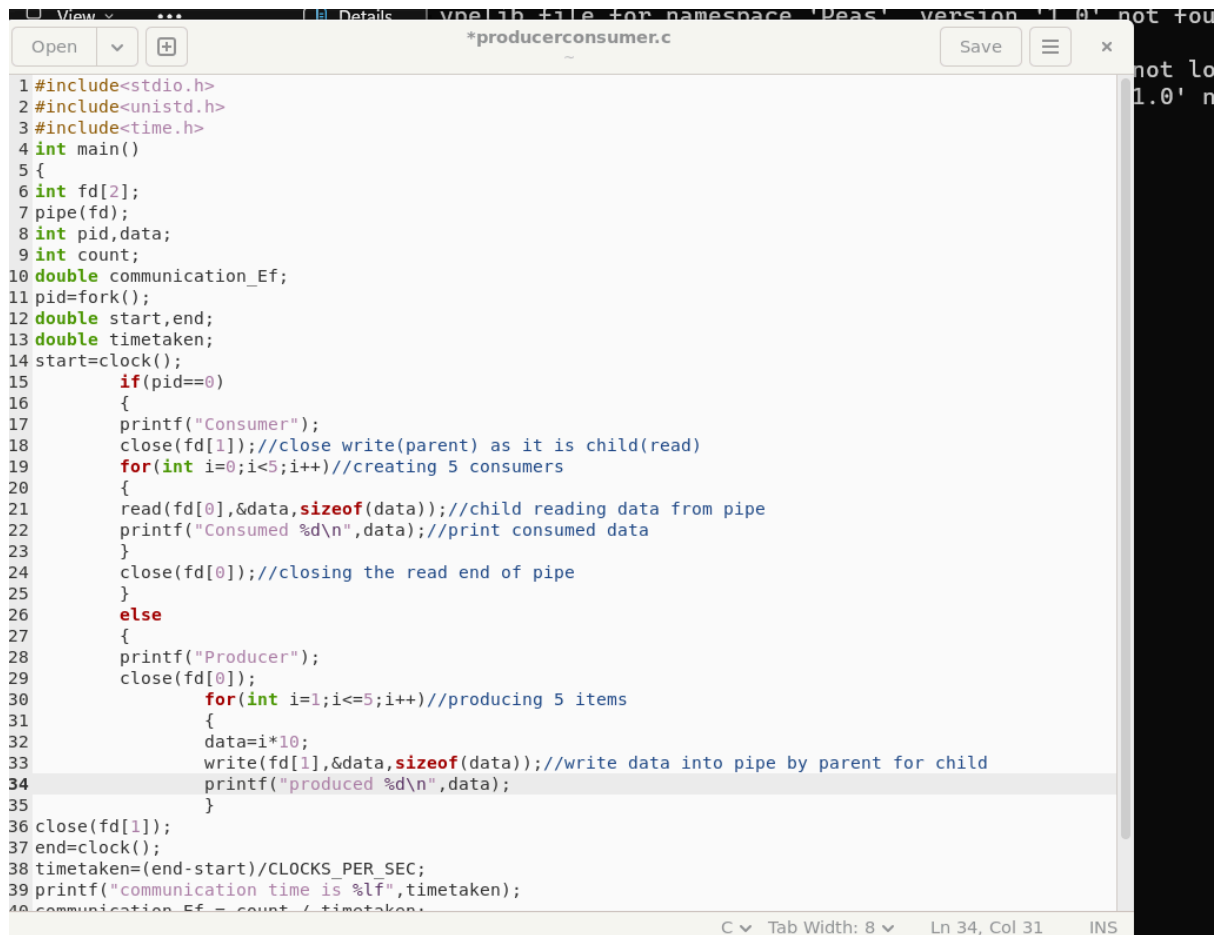


OSSP - PRACTICAL ASSIGNMENT - 5

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```
1 #include<stdio.h>
2 #include<unistd.h>
3 #include<time.h>
4 int main()
5 {
6     int fd[2];
7     pipe(fd);
8     int pid,data;
9     int count;
10    double communication_Ef;
11    pid=fork();
12    double start,end;
13    double timetaken;
14    start=clock();
15    if(pid==0)
16    {
17        printf("Consumer");
18        close(fd[1]); //close write(parent) as it is child(read)
19        for(int i=0;i<5;i++) //creating 5 consumers
20        {
21            read(fd[0],&data,sizeof(data)); //child reading data from pipe
22            printf("Consumed %d\n",data); //print consumed data
23        }
24        close(fd[0]); //closing the read end of pipe
25    }
26    else
27    {
28        printf("Producer");
29        close(fd[0]);
30        for(int i=1;i<=5;i++) //producing 5 items
31        {
32            data=i*10;
33            write(fd[1],&data,sizeof(data)); //write data into pipe by parent for child
34            printf("produced %d\n",data);
35        }
36        close(fd[1]);
37        end=clock();
38        timetaken=(end-start)/CLOCKS_PER_SEC;
39        printf("communication time is %lf",timetaken);
40        communication_Ef = count / timetaken;
```



```
37 end=clock();
38 timetaken=(end-start)/CLOCKS_PER_SEC;
39 printf("communication time is %lf",timetaken);
40 communication_Ef = count / timetaken;
41 printf("Communication Efficiency = %lf\n",communication_Ef);
42 }
43 }
```

```

: typelib file for namespace 'PeasGtk', version '1.0' not found
praneethgadila@praneethgadila:~$ gcc producerconsumer.c
praneethgadila@praneethgadila:~$ ./a.out
producerproduced 10
roduced 20
roduced 30
roduced 40
roduced 50
ommunication time is 0.000056Communication Efficiency = 0.000000
onsumerConsumed 10
onsumed 20
onsumed 30
onsumed 40
onsumed 50
praneethgadila@praneethgadila:~$ |

```

2.

```

1 #include<unistd.h>
2 #include<stdio.h>
3 int main()
4 {
5     int fd[2];
6     pipe(fd);
7     int pid=fork();
8     if(pid>0)
9     {
10        close(fd[0]);
11        dup2(fd[1],1);
12        execlp("ls","ls","-l",NULL);
13    }
14    if(pid==0)
15    {
16        close(fd[1]);
17        dup2(fd[0],0);
18        execlp("grep","grep",".c",NULL);
19    }
20 }

```

```

praneethgadila@praneethgadila:~$ gcc program5.c
praneethgadila@praneethgadila:~$ ./a.out
drwxr-xr-x 2 praneethgadila praneethgadila 4096 Jul 28 05:58 backup
-rwxr-xr-x 1 praneethgadila praneethgadila 16216 Sep 10 05:10 client
-rw-r--r-- 1 praneethgadila praneethgadila 278 Sep 10 04:56 client.c
-rwxr-xr-x 1 praneethgadila praneethgadila 16216 Sep 10 05:04 client1
-rw-r--r-- 1 praneethgadila praneethgadila 264 Sep 10 04:57 client1.c
-rw-r--r-- 1 praneethgadila praneethgadila 342 Aug 27 05:47 coding.c
drwxr-xr-x 2 praneethgadila praneethgadila 4096 Jul 28 05:58 config
-rw-r--r-- 1 praneethgadila praneethgadila 88 Jul 27 08:31 dd.c
drwxr-xr-x 2 praneethgadila praneethgadila 4096 Jul 28 05:58 docs
-rw-r--r-- 1 praneethgadila praneethgadila 97 Aug 11 04:53 exec1.c
-rw-r--r-- 1 praneethgadila praneethgadila 120 Aug 11 05:05 exec2.c
-rw-r--r-- 1 praneethgadila praneethgadila 0 Aug 11 05:05 exec3.
-rw-r--r-- 1 praneethgadila praneethgadila 107 Aug 11 05:11 exec3.c
-rw-r--r-- 1 praneethgadila praneethgadila 203 Aug 11 05:32 family.c
-rw-r--r-- 1 praneethgadila praneethgadila 247 Aug 11 06:04 family_fork.c
-rw-r--r-- 1 praneethgadila praneethgadila 404 Sep 1 05:50 fcfs.c
-rw-r--r-- 1 praneethgadila praneethgadila 293 Aug 4 05:44 fipe.c
-rw-r--r-- 1 praneethgadila praneethgadila 172 Jul 30 05:11 forkk.c
-rw-r--r-- 1 praneethgadila praneethgadila 117 Jul 27 08:36 gb.c
-rw-r--r-- 1 praneethgadila praneethgadila 259 Aug 11 06:06 grepp.c
-rw-r--r-- 1 praneethgadila praneethgadila 336 Aug 4 05:55 kiran.c
-rw-r--r-- 1 praneethgadila praneethgadila 240 Sep 15 05:51 malloc.c
-rw-r--r-- 1 praneethgadila praneethgadila 231 Sep 15 05:57 malloc1.c
-rw-r--r-- 1 praneethgadila praneethgadila 276 Sep 1 14:28 mids1.c
-rw-r--r-- 1 praneethgadila praneethgadila 301 Aug 27 05:39 namedpipex.c
-rw-r--r-- 1 praneethgadila praneethgadila 316 Jul 30 05:55 openn.c
-rw-r--r-- 1 praneethgadila praneethgadila 247 Aug 4 05:30 pipee.c
-rw-r--r-- 1 praneethgadila praneethgadila 578 Sep 10 05:20 posix.c
-rw-r--r-- 1 praneethgadila praneethgadila 320 Aug 20 05:35 practial1.c
-rw-r--r-- 1 praneethgadila praneethgadila 419 Aug 20 05:39 practial1.cx

```